

Zomato Restaurant Dataset — Analysis Notebook Documentation

Overview

This notebook performs data loading, cleaning, exploratory data analysis (EDA), and visualization on a Zomato restaurant dataset (test_zomato.csv, 9,552 rows × 22 columns). The workflow is organized into two parts: **Part 1** (data quality/structure) and **Part 2** (business insights/visualizations).

Environment

- **Language/Kernel:** Python 3.12.13
- **Libraries:** numpy, pandas, matplotlib, seaborn, os, warnings
- **Platform:** Kaggle (input path: /kaggle/input/datasets/adityapoddar8130/test-zomato)

Notebook Structure

Data Loading & Preparation (Cells 1–7)

Cell	Purpose
1	Import libraries, set plot style/warnings, list files in the dataset directory
2	Locate and load the CSV/XLSX file into df (encoding latin-1)
3	Fix unnamed index column → rename to ID (or create one), strip column name whitespace
4	Inspect shape (9,552 × 22), column names, and dtypes via df.info()
5	Compute missing value counts/percentages per column
6	Check and remove duplicate rows / duplicate Restaurant IDs (none found)
7	Clean/convert types: strip commas from Average Cost for two, convert Aggregate rating, Votes, Price range to numeric; fill missing Cuisines with "Not Specified"; strip City whitespace

Part 1 — Descriptive Statistics (Cells 8–9)

- Cell 8: df.describe() for numeric and categorical columns.
- Cell 9: Unique city count (142), top city by restaurant count (**New Delhi**, 5,473), most frequent cuisine (**North Indian**, 3,960).

Part 2 — Business Insights & Visualizations (Cells 10–20)

Cell	Insight / Chart
10	Core metrics: average rating (all vs. rated-only), average/median cost for two
11	Restaurant with highest votes (Toit , Bangalore) + top 10 most-voted restaurants
12	Average rating by online delivery availability
13	Average rating by table booking availability
14	Top 10 cities by restaurant count
15	Bar chart: Top 10 cities by restaurant count
16	Histogram: Distribution of aggregate ratings
17	Bar chart: Restaurants offering online delivery
18	Bar chart: Top 10 cuisines
19	Line chart: Average rating by price range
20	Heatmap: Correlation between numeric features (cost, price range, rating, votes)

Key Findings

- **Dataset:** 9,552 restaurants, 142 cities, 1,827 unique cuisine combinations.
- **Missing data:** Minimal ($\leq 0.1\%$ per column); Cuisines had 10 missing values, filled as "Not Specified".
- **No duplicates** found in restaurant IDs or full rows.
- **Ratings:** Mean rating (excluding unrated) = 3.44; higher price ranges and table booking correlate with higher average ratings.
- **Cost:** Median cost for two = ₹400 (mean skewed high by outliers up to ₹800,000).
- **Geography:** New Delhi dominates restaurant count; North Indian is the most common cuisine.

Reproducibility Notes

- Data path is hardcoded to a Kaggle input directory — will need adjustment to run elsewhere.
- Random/inference logic is absent (purely descriptive EDA, no modeling).